SCADA

What is SCADA?

Supervisory **C**ontrol **A**nd **D**ata **A**cquisition

used to monitor and control a plant or equipment in industries such as telecommunications, water and waste control, energy, oil & gas, refining and transportation.

- SCADA is not a specific technology, but a type of application.

- Any application that gets data about a system in order to control that system is a SCADA application.

SCADA Functions

1. Data acquisition
2. Networked data communication
3. Data presentation
4. Control

SCADA Components

Sensors and Actuators

- Sensors convert input parameters of the managed system (like temperature or angle of rotation) to measurable digital or analog electrical signals.
- Actuators (e.g. control relays) generate proper signals to perform the control function.

Remote telemetry units (RTUs)/PLCs

- These are small computerized units deployed in the field at specific sites and locations.
- RTUs serve as local collection points for gathering data from sensors and delivering commands to actuators.

SCADA Components

Host Computer

- A central computer server or servers sometimes called a SCADA Center, master station, or Master Terminal Unit (MTU) with large interactive terminals for display and control which act as human interface.
- The central host computer or master station is most often a single computer or a network of computer servers that provide a man-machine operator interface to the SCADA system.
- Operator terminals are connected to the central host computer by a LAN/WAN so that the viewing screens and associated data can be displayed for the operators.

SCADA Components

Host Computer

The central host computer acts as a server for the SCADA application, and the operator terminals are clients that request and send information to the central host computer based on the request and action of the operators.

SCADA	HMI
SCADA software installed on computers or workstations or servers	HMI is available in touch panels or stand-alone simple hardware with the HMI software
SCADA acts as a graphical user interface between PLC and operators	HMI also acts as a direct interface between PLC and operators but with limited functionality
SCADA used for remote monitoring and control of industrial processes	HMI mainly used for local monitoring and control of single industrial process.
SCADA handles large amounts of data and displays it in a structured way	HMI is Limited in terms of data handling and display capabilities
SCADA can be programmed to perform advanced control strategies	HMI is limited in terms of advanced control capabilities
SCADA can be integrated with other systems, such as PLC and DCS	HMI is typically standalone and not as easily integrated with other systems or with limited functions

SCADA can be programmed to create alarms and notifications for critical events via different communication protocols	**HMI is limited in terms of alarm and notification capabilities as compared to SCADA**
SCADA can be used for long-term data logging and storing historical data of process control instruments	HMI is limited in terms of data logging and storage capabilities
SCADA provides trend analysis and reporting of data in custom templates as per the user's requirement	HMI is limited in terms of data analysis and reporting capabilities
SCADA can be configured for access control and security to protect against unauthorized access	HMI does not have advanced security features

SCADA Components

Communications Network

- It is intended to provide the means by which data can be transferred between the central host computer servers and the field-based RTUs.

- The Communication Network refers to the equipment needed to transfer data to and from different sites. The medium used can either be cable, telephone or radio.

SCADA Protocols

IEC

- International Electrotechnical Commission (IEC) is a non-profit non-govt. international standards organization that sets standards and protocols for a very wide range of Electric and Electronic devices and communications.
- Some IEC standards related to SCADA are,…

SCADA Protocols

MODBUS

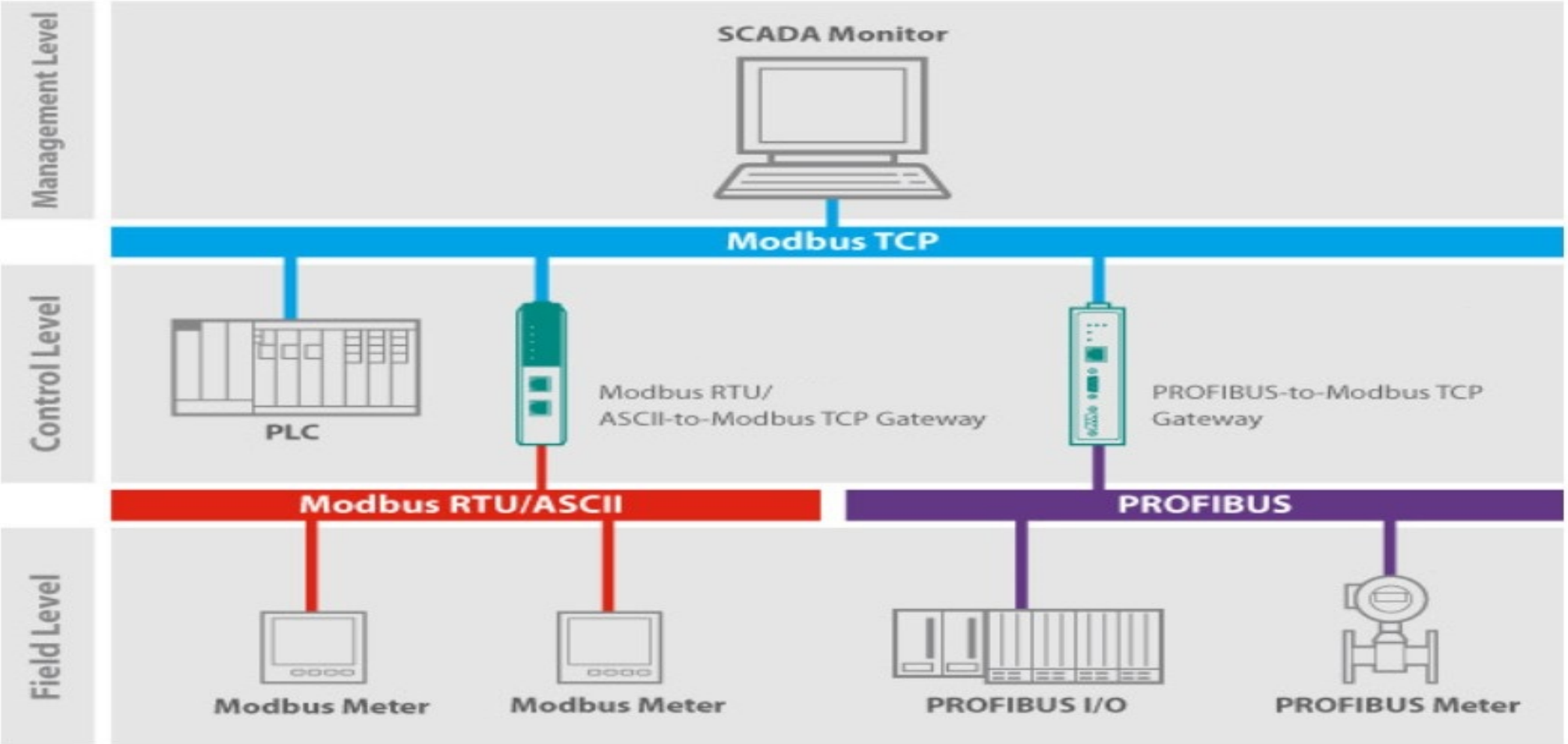
- Most popular of the communication protocols used in SCADA, MODBUS is an application layer (Level 7 of OSI model) messaging protocol that provides Master-Slave/Client-Server communication between devices connected on different types of buses or networks.

- In simple terms, it is a method used for transmitting information over serial lines between electronic devices. In SCADA, it is used in communication between Host Station and Controller (RTU/PLC/PAC), where Host Station is the Master. It is also used in communication between Controller and Field Instrument, where Controller is the Master.

- Originally developed for only serial messaging, the protocol has been expanded to include implementations over Serial, TCP/IP and the User Datagram Protocol (UDP).

SCADA Protocols

MODBUS



SCADA Protocols

IEEE 802.3 Ethernet

The term Ethernet refers to the family of local-area network (LAN) products covered by the IEEE 802.3 standard that defines what is commonly known as the CSMA/CD protocol. Three data rates are currently defined for operation over optical fiber and twisted-pair cables:

- 10 Mbps 10Base-T Ethernet
- 100 Mbps Fast Ethernet
- 1000 Mbps Gigabit Ethernet
- Is easy to understand, implement, manage, and maintain
- Allows low-cost network implementations
- Provides extensive topological flexibility for network installation
- Guarantees successful interconnection and operation of
- standards- compliant products, regardless of manufacturer

SCADA Protocols

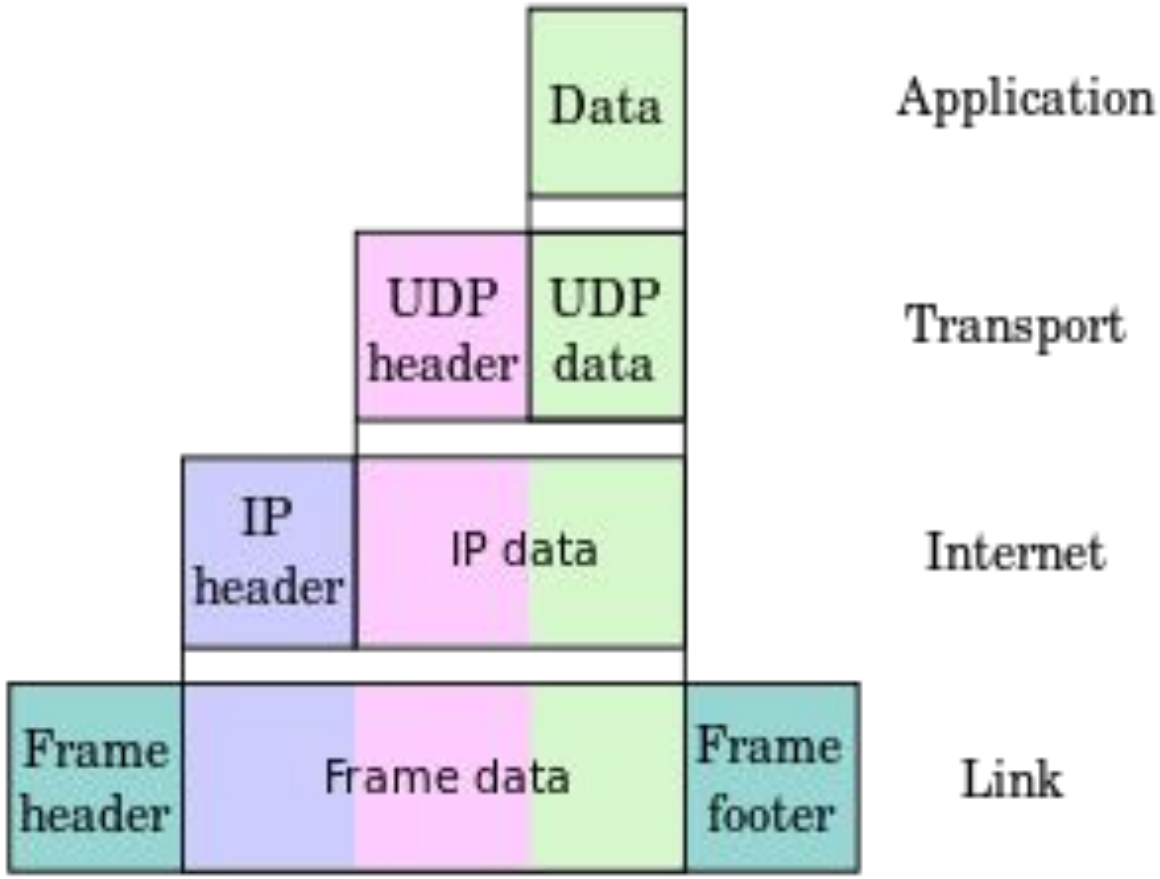
TCP/IP

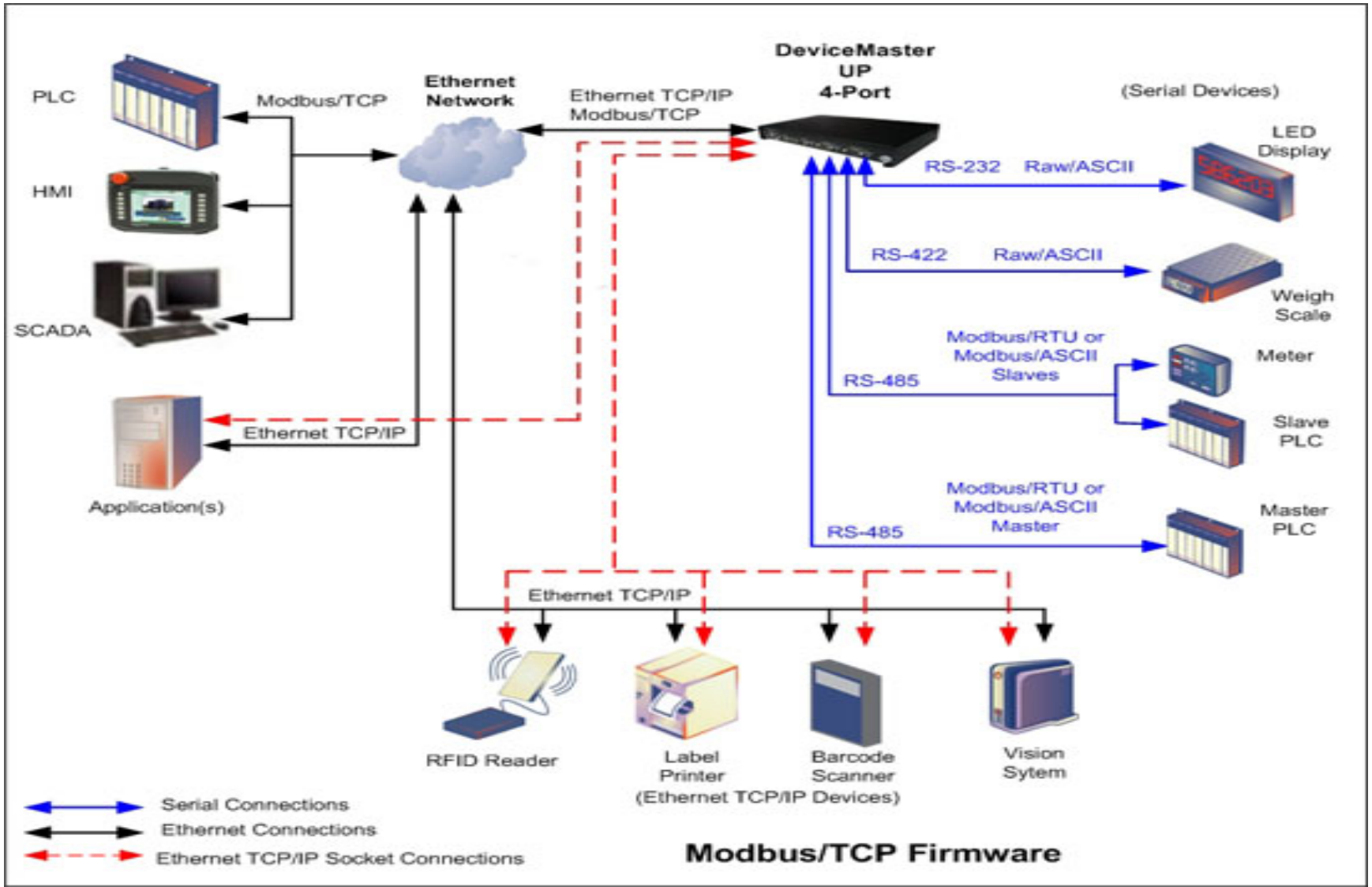
- **TCP/IP** stands for Transmission Control Protocol/Internet Protocol, which is a set of networking protocols that allows two or more computers to communicate.

- Since all modern controllers and many field instruments have built-in Microprocessors/Microcontrollers, TCP/IP is widely used in today's SCADA applications, particularly for remote control and monitoring over wide area network.

SCADA Protocols

TCP/IP



SCADA Protocols

TCP/IP